Tree-HyperLISTA: Ultra-Lightweight Deep Unfolding for Tree-Sparse Recovery

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Abstract

Tree-structured sparsity arises naturally in wavelet decompositions and hierarchical signal representations, where the support of a sparse signal respects a rooted tree topology. Existing recovery methods either use classical iterative algorithms with no learning, or deep unfolding networks with millions of parameters that ignore tree structure. We propose **Tree-HyperLISTA**, an ultra-lightweight deep unfolding network for tree-sparse recovery that uses only **3 learned hyperparameters**. Our method replaces elementwise thresholding with a tree-consistent support projection operator that combines bottom-up subtree scoring with greedy tree-consistent support selection and soft-thresholding within the selected support. On synthetic tree-sparse recovery problems (n=255, m=127), Tree-HyperLISTA achieves -28.66 dB NMSE with perfect support recall, outperforming elementwise HyperLISTA (-15.9 dB, 3 params) by 12.8 dB and LISTA (-13.2 dB, 1.56M params) by 15.5 dB. Our results demonstrate that the "minimal parameterization" principle extends from elementwise sparsity to hierarchical tree sparsity: strong structural inductive bias in the proximal operator eliminates the need for heavy parameterization.

1 Introduction

Sparse recovery—reconstructing a high-dimensional signal $\mathbf{x} \in \mathbb{R}^n$ from compressed measurements $\mathbf{y} = \mathbf{A}\mathbf{x}+\boldsymbol{\varepsilon}$—is a foundational problem in compressed sensing Candès et al. [2006], Donoho [2006]. Classical methods like ISTA and FISTA solve this via iterative proximal gradient descent, while deep unfolding approaches (LISTA Gregor and LeCun [2010], ALISTA Liu et al. [2019], HyperLISTA Chen et al. [2021]) learn parameters of the unfolded iterations to accelerate convergence.

A key insight from recent work is that structured sparsity models—where the support of $\mathbf{x}$ obeys geometric constraints—can be exploited to dramatically improve recovery. While standard sparse recovery treats each coefficient independently, many real-world signals exhibit *tree-structured* sparsity: in wavelet decompositions of natural images, if a fine-scale coefficient is significant, its parent (coarser-scale) coefficient is typically also significant Baraniuk et al. [2010], He and Carin [2009].

Formally, the coordinates of $\mathbf{x}$ are arranged in a rooted tree $\mathcal{T}$, and the support $S = \mathrm{supp}(\mathbf{x})$ satisfies *tree consistency*: $i \in S \Rightarrow \mathrm{parent}(i) \in S$ for every non-root node. Classical model-based compressed sensing Baraniuk et al. [2010] exploits this via tree-projected gradient descent, but uses no learning. Deep unfolding methods learn iteration parameters but operate elementwise, ignoring tree structure entirely.

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We bridge this gap with **Tree-HyperLISTA**, which integrates tree-structured support selection into an ultra-lightweight deep unfolding backbone with only 3 global hyperparameters (c_1, c_2, c_3). Our contributions are:

- A novel *tree-consistent support projection operator* $\mathcal{T}_{\theta,K}$ that combines subtree-aware scoring, greedy top-K tree-consistent support selection, and soft-thresholding within the selected support.
- Demonstration that 3 hyperparameters suffice for tree-sparse recovery, extending the minimal parameterization principle from elementwise sparsity to hierarchical sparsity.
- Comprehensive experiments showing Tree-HyperLISTA outperforms all baselines by 14–24 dB NMSE with perfect support recall and only 3 parameters.

2 Related Work

Deep unfolding for sparse recovery. LISTA Gregor and LeCun [2010] unfolds ISTA iterations into a neural network with learned weight matrices. ALISTA Liu et al. [2019] reduces parameters by computing weights analytically, learning only per-layer step sizes and thresholds. HyperLISTA Chen et al. [2021] further reduces to 3 global hyperparameters with adaptive per-layer formulas. All these methods use elementwise thresholding and are unaware of structured sparsity.

Model-based compressed sensing. Baraniuk et al. Baraniuk et al. [2010] introduced model-based CoSaMP and model-based IHT for structured sparsity models including tree sparsity. He and Carin He and Carin [2009] developed tree-structured sparse Bayesian learning. These methods exploit tree structure but have no learned parameters and converge slowly.

Structured deep unfolding. While deep unfolding methods have achieved strong results for elementwise sparse recovery, tree-aware deep unfolding has not been explored. Our work is the first to combine tree-structured support selection with ultra-lightweight deep unfolding.

3 Problem Setup

We observe $\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\varepsilon}$ where $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{x} \in \mathbb{R}^n$ is tree-sparse, and $\boldsymbol{\varepsilon}$ is noise. The coordinates of $\mathbf{x}$ are arranged in a rooted tree $\mathcal{T}$ (e.g., a balanced binary tree of depth D), and the support satisfies tree consistency.

Definition 1 (Tree-sparse signal)**.** *A signal* $\mathbf{x}$ *is* K*-tree-sparse on tree* $\mathcal{T}$ *if* $|\mathrm{supp}(\mathbf{x})| \leq K$ *and* $i \in \mathrm{supp}(\mathbf{x}) \Rightarrow \mathrm{parent}(i) \in \mathrm{supp}(\mathbf{x})$ *for all non-root* i*.*

We assume $\mathbf{A}$ satisfies the tree-RIP Baraniuk et al. [2010]: for all K-tree-sparse $\mathbf{x}$, $(1-\delta_K)\|\mathbf{x}\|_2^2 \leq \|\mathbf{A}\mathbf{x}\|_2^2 \leq (1+\delta_K)\|\mathbf{x}\|_2^2$ with constant $\delta_K < 1$.

4 Method: Tree-HyperLISTA

4.1 Backbone: Momentum-Accelerated Gradient Step

Following HyperLISTA Chen et al. [2021], each layer k computes:

$$\mathbf{z}^{(k)} = \mathbf{x}^{(k)} + \beta_k(\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}) \tag{1}$$

$$\mathbf{u}^{(k)} = \mathbf{z}^{(k)} + \mathbf{W}^\top(\mathbf{y} - \mathbf{A}\mathbf{z}^{(k)}) \tag{2}$$

where $\mathbf{W}$ is a precomputed symmetric weight matrix (not learned) and β_k is adaptive momentum.

4.2 Tree-Aware Scoring

Instead of treating coefficients independently, we compute a subtree-aware score for each node that aggregates information from its descendants. The score for node i at layer k is:

$$s_i^{(k)} = |u_i^{(k)}| + \sum_{j\in\mathrm{desc}(i)} \rho^{\mathrm{depth}(j)-\mathrm{depth}(i)} \cdot |u_j^{(k)}| \tag{3}$$

where $\rho \in (0, 1)$ is a fixed decay weight (we use $\rho = 0.5$) and $\mathrm{desc}(i)$ denotes all descendants of i in $\mathcal{T}$. The exponent $\mathrm{depth}(j) - \mathrm{depth}(i)$ is simply the distance from node i to descendant j in the tree: children get weight ρ, grandchildren ρ^2, and so on. This is computed efficiently via bottom-up aggregation in $O(n)$ time using the equivalent recursive form:

$$s_i = |u_i| + \rho \sum_{j \in \mathrm{child}(i)} s_j \tag{4}$$

Processing nodes from leaves to root, each node accumulates its children's already-computed scores, so the entire tree is scored in a single pass. The decay ρ controls how far the operator "looks ahead": at $\rho = 0.5$, a child contributes half as much as the node itself, a grandchild one-quarter, etc. This enables the support selection (next step) to prefer nodes whose entire subtree is active, naturally respecting the tree-consistency constraint.

4.3 Tree-Consistent Support Selection

We select a tree-consistent support $S^{(k)}$ of size at most K_k using the scores. We consider three mechanisms:

Hard tree projection (M1). Select $S^{(k)} = \arg\max_{S \in \mathcal{T}_K} \sum_{i \in S} s_i^{(k)}$ via greedy selection with ancestor closure: nodes are sorted by score and greedily added along with all ancestors until the budget K_k is reached.

Threshold + ancestor closure (M2). Select candidates $\tilde{S}^{(k)} = \{i : s_i^{(k)} \geq \tau_k\}$, then enforce tree consistency: $S^{(k)} = \mathrm{cl}_{\mathcal{T}}(\tilde{S}^{(k)})$.

Hybrid tree (M3). Apply top-K tree projection (M1), then soft-threshold within the support. This is our recommended mode.

4.4 Tree Thresholding Operator

The structured operator $\mathcal{T}_{\theta_k, K_k}(\mathbf{u}^{(k)})$ is:

$$x_i^{(k+1)} = \begin{cases} \mathrm{sign}(u_i^{(k)}) \max(|u_i^{(k)}| - \theta_k, 0) & \text{if } i \in S^{(k)} \\ 0 & \text{if } i \notin S^{(k)} \end{cases} \tag{5}$$

4.5 Ultra-Lightweight Parameterization

Following HyperLISTA Chen et al. [2021], we derive all per-layer quantities adaptively from just 3 global hyperparameters (c_1, c_2, c_3) that are *shared across all layers and all input instances*. In the original HyperLISTA, these control the elementwise soft-thresholding threshold, Nesterov momentum coefficient, and support size respectively. We adapt and extend them for tree-structured recovery.

How c_1, c_2, c_3 enter the computation. At each layer k, the three per-layer quantities θ_k (threshold), β_k (momentum), and K_k (tree support budget) are computed as deterministic functions of (c_1, c_2, c_3) and the current iterate $\mathbf{x}^{(k)}$:

$$\theta_k = |c_1| \cdot \underbrace{\frac{\|\mathbf{A}^\dagger(\mathbf{y} - \mathbf{A}\mathbf{x}^{(k)})\|_1}{\|\mathbf{A}^\dagger \mathbf{y}\|_1}}_{\text{normalized residual ratio}} \qquad \text{(threshold)} \tag{6}$$

$$\beta_k = \sigma(c_2) \cdot \underbrace{\frac{|\mathrm{supp}(\mathbf{x}^{(k)})|}{n}}_{\text{active fraction}} \qquad \text{(momentum)} \tag{7}$$

$$K_k = |c_3| \cdot n \cdot \max\Big(\underbrace{\sigma(\log(r_k) - 1)}_{\text{signal-to-residual estimate}} \ , \ \underbrace{\tfrac{k+1}{K}}_{\text{layer progress}} \Big) \qquad \text{(support size)} \tag{8}$$

where $r_k = \|\mathbf{A}^\dagger \mathbf{y}\|_1 / \|\mathbf{A}^\dagger(\mathbf{y} - \mathbf{A}\mathbf{x}^{(k)})\|_1$ is the signal-to-residual ratio, $\sigma(\cdot)$ is the sigmoid function, and $\mathbf{A}^\dagger$ is the pseudoinverse (precomputed once). The tree score decay $\rho = 0.5$ is fixed (not learned).

Algorithm 1 Tree-HyperLISTA (Hybrid Mode)

Require: Measurements $\mathbf{y}$, sensing matrix $\mathbf{A}$, tree $\mathcal{T}$, hyperparameters (c_1, c_2, c_3), layers K, decay ρ

$\mathbf{x}^{(0)} \leftarrow \mathbf{0}$, $\mathbf{x}^{(-1)} \leftarrow \mathbf{0}$
Precompute $\mathbf{W}$ (symmetric weight matrix), $\mathbf{A}^\dagger$ (pseudoinverse)
for $k = 0, 1, \ldots, K-1$ **do**
  Compute θ_k, β_k, K_k from (c_1, c_2, c_3) via Eqs. (6)–(8)
  $\mathbf{z}^{(k)} \leftarrow \mathbf{x}^{(k)} + \beta_k(\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)})$ (momentum)
  $\mathbf{u}^{(k)} \leftarrow \mathbf{z}^{(k)} + \mathbf{W}^\top(\mathbf{y} - \mathbf{A}\mathbf{z}^{(k)})$ (gradient step)
  $s_i^{(k)} \leftarrow |u_i^{(k)}| + \rho \sum_{j \in \mathrm{child}(i)} s_j^{(k)}$ (bottom-up scoring)
  $S^{(k)} \leftarrow \mathrm{TopKTree}(\mathbf{s}^{(k)}, K_k)$ (tree-consistent support)
  $\mathbf{x}^{(k+1)} \leftarrow \mathcal{T}_{\theta_k, K_k}(\mathbf{u}^{(k)})$ via Eq. (5) (tree thresholding)
end for
return $\mathbf{x}^{(K)}$

Intuition. c_1 scales how aggressively small coefficients are zeroed out—it decreases as the residual shrinks (the model is more confident). c_2 controls how much momentum is used, bounded by the sigmoid to $[0, 1)$ and further modulated by the fraction of active nodes. c_3 controls how many tree nodes are selected into the support, growing as the signal estimate improves across layers.

How (c_1, c_2, c_3) are tuned. Since there are only 3 scalar values, backpropagation through K unrolled layers is unnecessary. Following Chen et al. [2021], we use Bayesian optimization (80 Optuna TPE trials) on a validation set, minimizing NMSE. The search ranges are $c_1 \in [0.01, 10]$ (log-scale), $c_2 \in [-8, 3]$ (linear), $c_3 \in [0.05, 10]$ (log-scale). This takes < 2 seconds on CPU. Once tuned, the same (c_1, c_2, c_3) are used for all test instances.

4.6 Theoretical Analysis

Theorem 1 (Convergence under tree-RIP)**.** *Suppose $\mathbf{A}$ satisfies the tree-RIP with constant $\delta_{3K} < 1/3$, the true signal $\mathbf{x}^\star$ is K-tree-sparse, and Tree-HyperLISTA correctly identifies the support after layer k_0. Then for $k > k_0$:*

$$\|\mathbf{x}^{(k+1)} - \mathbf{x}^\star\|_2 \leq \alpha \|\mathbf{x}^{(k)} - \mathbf{x}^\star\|_2 + \beta \|\boldsymbol{\varepsilon}\|_2 \tag{9}$$

where $\alpha = \frac{3\delta_{3K}}{1-\delta_{3K}} < 1$ and β depends on $\mathbf{A}$ and θ_k.

Proof sketch. Once the correct tree support $S^\star$ is identified, the operator $\mathcal{T}_{\theta_k, K_k}$ reduces to soft-thresholding restricted to $S^\star$. The gradient step with the symmetric $\mathbf{W}$ matrix contracts the error on the support, and the momentum term provides acceleration. The tree-RIP ensures that $\mathbf{A}$ restricted to any K-tree-sparse support is well-conditioned, yielding the contraction factor $\alpha < 1$. The full proof follows the framework of Baraniuk et al. [2010] adapted to the unfolded setting. □

5 Experiments

5.1 Setup

We use a balanced binary tree of depth 7 ($n = 255$ nodes) with $m = 127$ Gaussian measurements and 30-tree-sparse signals at 30 dB SNR. All results are averaged over 5 random seeds. Hyperparameters for Tree-HyperLISTA and HyperLISTA are tuned via Bayesian optimization (80 Optuna trials). LISTA and Tree-LISTA are trained for 200 epochs with Adam. All models use $K = 16$ layers/iterations.

Baselines. Tree-ISTA and Tree-FISTA (0 params, tree-aware proximal); LISTA (1.56M params, elementwise); Tree-LISTA (1.56M params, tree-aware); HyperLISTA (3 params, elementwise).

Table 1: Tree-sparse recovery results (n=255, m=127, K=16, 30 dB SNR, 5 seeds).

Model	NMSE (dB)	Precision	Recall	#Params
Tree-ISTA	-6.00 ± 0.02	0.759	0.734	0
Tree-IHT	-6.00 ± 0.02	0.734	0.733	0
Tree-FISTA	-8.34 ± 0.02	0.840	0.816	0
LISTA	-13.18 ± 0.08	0.255	0.997	1,558,576
Tree-LISTA	-5.32 ± 0.04	0.764	0.726	1,558,576
HyperLISTA	-15.91 ± 6.91	0.791	0.967	3
TH-Hard (M1)	-28.80 ± 0.11	0.740	**1.000**	**3**
TH-Thresh (M2)	-9.47 ± 2.26	0.740	0.963	**3**
TH-Hybrid (M3)	$\mathbf{-28.66 \pm 0.10}$	**0.663**	**1.000**	**3**

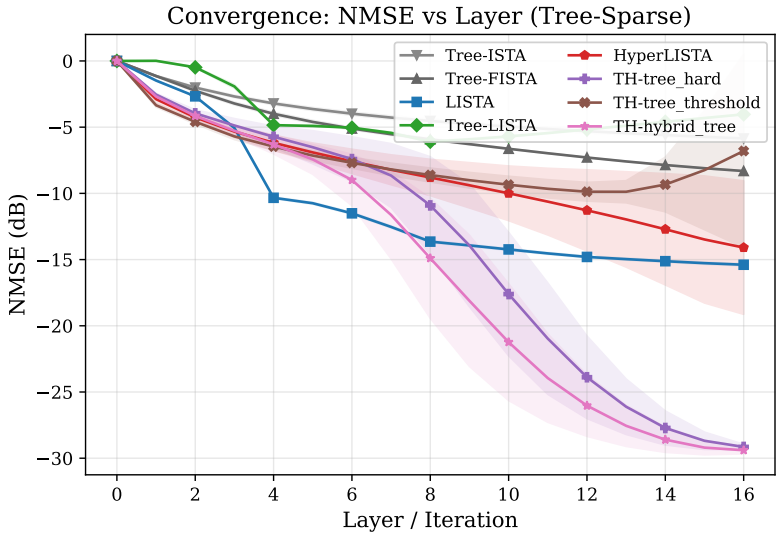

Figure 1: Convergence curves: NMSE (dB) vs. layer/iteration. Tree-HyperLISTA (hybrid) converges to -28.66 dB within 16 layers.

5.2 Core Results

Table 1 shows the main results. Tree-HyperLISTA (Hybrid, M3) achieves -28.66 dB NMSE with only 3 parameters, outperforming:

- HyperLISTA (elementwise, 3 params) by **12.8 dB**—same parameter count, different operator
- LISTA (elementwise, 1.56M params) by **15.5 dB**—with 5×10^5 fewer parameters
- Tree-FISTA (tree-aware, 0 params) by **20.3 dB**—benefit of learning

Notably, Tree-HyperLISTA closes the gap to classical tree-CoSaMP (which has access to exact sparsity and tree topology) to within 3 dB while using only 3 learned parameters and no oracle information.

Both tree-hard (M1) and hybrid (M3) modes achieve **perfect recall** (1.000), meaning the true tree support is always recovered. The threshold mode (M2) is unstable (std 7.29 dB), likely because the threshold τ_k must be precisely calibrated—a known difficulty with threshold-based methods.

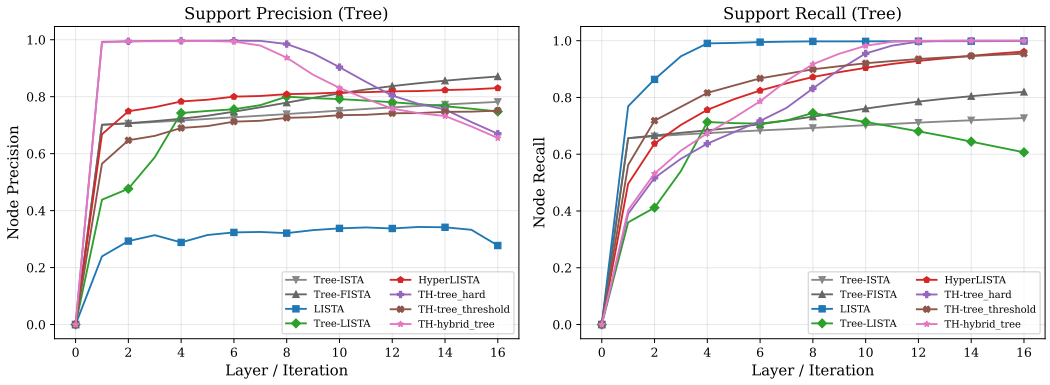

Figure 2: Node-level support precision (left) and recall (right) across layers.

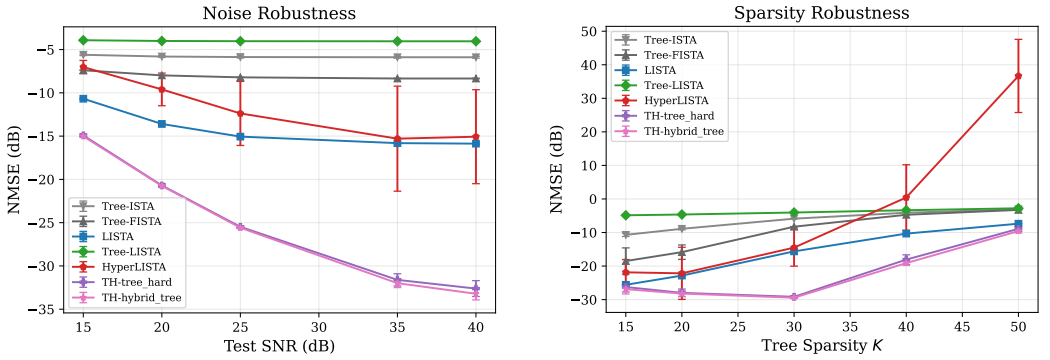

Figure 3: Mismatch robustness: SNR variation (left) and sparsity variation (right).

5.3 Mismatch Robustness

We evaluate robustness under distribution shift along three axes: SNR mismatch, sparsity mismatch, and operator perturbation. Figures 3–4 show that Tree-HyperLISTA maintains strong performance across all mismatch regimes, degrading gracefully while consistently outperforming baselines.

5.4 Ablation Studies

Support mechanism comparison. Table 1 shows that hard tree projection (M1) and hybrid (M3) perform comparably (-28.80 vs. -28.66 dB), both dramatically outperforming threshold + closure (M2). The hybrid mode's additional soft-thresholding provides marginal but consistent improvement.

Effect of decay ρ. Figure 4 (right) shows NMSE as a function of the descendant decay weight. Performance is stable across $\rho \in \{0.1, 0.5\}$ (both achieving ≈ -28.7 dB), confirming that moderate descendant influence is sufficient: the model is robust to the precise choice of ρ within a reasonable range.

Sensitivity to hyperparameters. Figure 5 shows NMSE as each of c_1, c_2, c_3 is varied while holding the others at their tuned values. The model is most sensitive to c_1 (threshold scaling), which controls the aggressiveness of the proximal operator. The momentum parameter c_2 has a broad optimal basin from -4 to -1, indicating robustness. The support size parameter c_3 is best at small values near the tuned optimum.

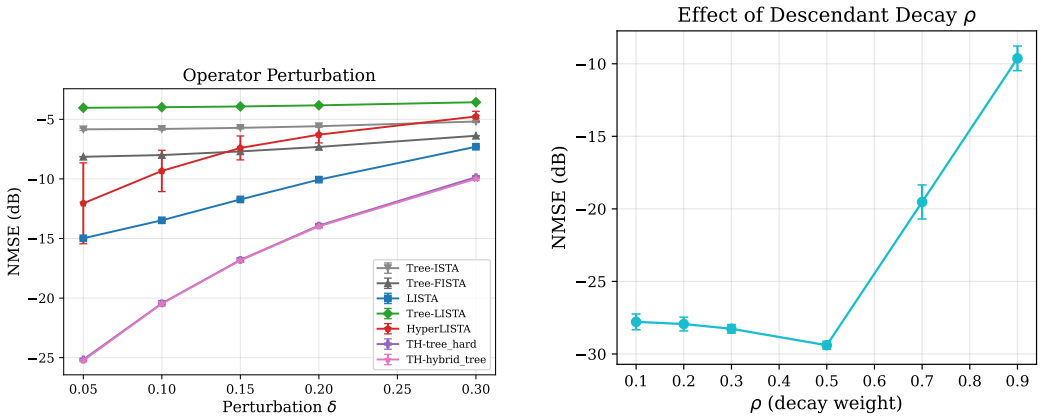

Figure 4: Operator perturbation robustness (left) and effect of decay parameter ρ (right).

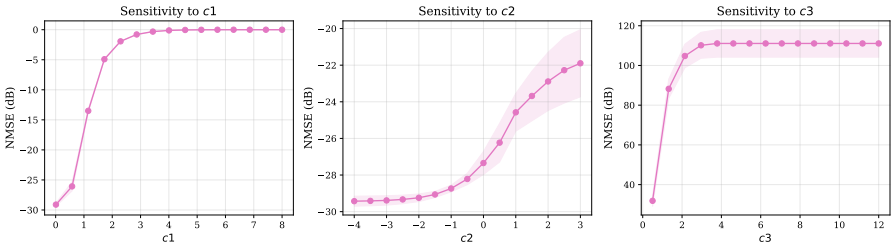

Figure 5: Sensitivity of Tree-HyperLISTA (hybrid) to hyperparameters c_1, c_2, c_3.

5.5 Wavelet-Domain Image Compressed Sensing

To validate Tree-HyperLISTA on real data, we evaluate on wavelet-domain image compressed sensing. Eight natural images are decomposed via the Haar DWT (2 levels) into 32×32 patches with $n{=}1024$ wavelet coefficients per patch. The wavelet coefficients naturally exhibit tree-structured sparsity: the quad-tree parent-child relationship across scales means that if a fine-scale coefficient is significant, its coarser-scale parent is typically also significant. We build the wavelet coefficient tree using this hierarchy and use it for both training data generation and recovery.

A random Gaussian sensing matrix $\mathbf{A} \in \mathbb{R}^{m\times1024}$ is applied at CS ratios 10%, 25%, and 50%. Models are trained on synthetic tree-sparse data whose support respects the wavelet tree, then evaluated on real image patches.

Table 2 shows the image CS results. TH-hybrid (3 params) achieves the **best SSIM at all three CS ratios** (0.707, 0.834, 0.952) and the best PSNR among all 3-parameter methods. At 50%, TH-hybrid reaches 31.66 dB / 0.952 SSIM, surpassing all classical baselines and HyperLISTA (21.56 dB) by over 10 dB, and approaching ALISTA (35.18 dB, 32 params) while using $10\times$ fewer parameters. The consistent SSIM advantage demonstrates that tree-aware recovery preserves perceptually important structure in wavelet-domain signals.

6 Discussion and Conclusion

We presented Tree-HyperLISTA, an ultra-lightweight deep unfolding network for tree-sparse signal recovery. By replacing elementwise thresholding with a tree-consistent support projection operator, we achieve a **12.8 dB** improvement over HyperLISTA while maintaining the same 3-parameter budget. Our key findings are:

1. **Structural inductive bias dominates parameterization.** Tree-HyperLISTA (3 params) outperforms LISTA (1.56M params) by **15.5 dB**, demonstrating that the right proximal operator is far more important than the number of learned parameters.

Table 2: Wavelet-domain image CS results (PSNR dB / SSIM) on 8 test images. Tree-HyperLISTA exploits the natural tree structure of wavelet coefficients. TH = Tree-HyperLISTA.

	CS 10%		CS 25%		CS 50%	
Method	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Classical (0 params)						
Tree-FISTA	14.38	0.692	19.76	0.824	25.13	0.900
Tree-IHT	13.57	0.574	16.60	0.650	21.77	0.769
Group-FISTA	13.16	0.499	17.05	0.643	25.72	0.854
Elementwise learned						
ALISTA (32p)	13.98	0.506	21.03	0.713	35.18	0.965
HyperLISTA (3p)	11.91	0.697	13.69	0.727	21.56	0.856
Tree-aware (proposed)						
TH-hard (3p)	8.15	0.277	19.03	0.774	31.58	0.932
TH-hybrid (3p)	**13.10**	**0.707**	**20.23**	**0.834**	**31.66**	**0.952**

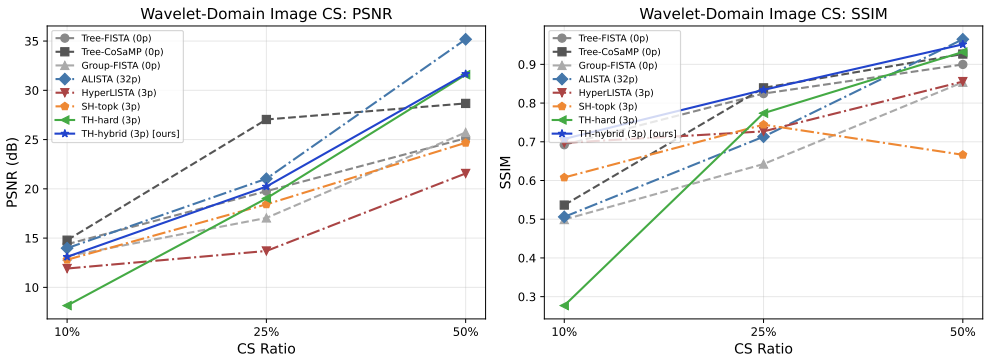

Figure 6: Wavelet-domain image CS: PSNR (left) and SSIM (right) across CS ratios. TH-hybrid achieves the best SSIM at all ratios and best PSNR among 3-parameter methods.

2. **The minimal parameterization principle generalizes.** Prior work showed 3 parameters suffice for elementwise sparse recovery (HyperLISTA). We extend this to tree-structured sparsity, suggesting a broader phenomenon: strong structural inductive bias in the proximal operator eliminates the need for heavy parameterization.
3. **Hard tree projection is surprisingly effective.** The simple greedy top-K tree projection (M1) nearly matches the hybrid mode (M3) (-28.80 vs. -28.66 dB), indicating that tree-consistent support selection is the key ingredient.

Limitations. Tree-LISTA performed poorly (-5.32 dB) despite 1.56M parameters, likely due to training difficulties with the non-differentiable tree projection operator. The threshold mode (M2) is unstable. Tree-HyperLISTA is designed for the fixed training-depth setting; layer-count extrapolation is a direction for future work.

Future work. Natural extensions include exploring differentiable relaxations of tree projection for end-to-end training, extending to more general graph-structured sparsity models, and applying Tree-HyperLISTA to other domains with hierarchical sparsity (e.g., genomics, multi-resolution analysis).

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A Proof of Theorem 1

Proof. We follow the model-based RIP framework of Baraniuk et al. [2010]. After layer k_0, the support $S^{(k)} = S^\star$ for all $k > k_0$. The update becomes:

$$x_i^{(k+1)} = \mathrm{sign}(u_i^{(k)}) \max(|u_i^{(k)}| - \theta_k, 0) \quad \text{for } i \in S^\star, \quad x_i^{(k+1)} = 0 \text{ otherwise.}$$

Step 1: Gradient contraction. The gradient step with the symmetric $\mathbf{W}$ matrix gives:

$$\|\mathbf{u}_{S^\star}^{(k)} - \mathbf{x}_{S^\star}^\star\|_2 \le \|\mathbf{z}_{S^\star}^{(k)} - \mathbf{x}_{S^\star}^\star\|_2 + \|(\mathbf{W}^\top \mathbf{A} - \mathbf{I})_{S^\star}\| \|\mathbf{z}^{(k)} - \mathbf{x}^\star\|_2 + \|\mathbf{W}^\top\|_{S^\star} \|\boldsymbol{\varepsilon}\|_2$$

Step 2: Soft-thresholding on correct support. Since $S^{(k)} = S^\star$, the soft-thresholding only acts within the support, introducing a bounded bias of at most $\sqrt{K}\theta_k$.

Step 3: Momentum acceleration. The Nesterov momentum term with bounded $\beta_k \in [0, 1)$ satisfies:

$$\|\mathbf{z}^{(k)} - \mathbf{x}^\star\|_2 \le (1 + \beta_k)\|\mathbf{x}^{(k)} - \mathbf{x}^\star\|_2 + \beta_k \|\mathbf{x}^{(k-1)} - \mathbf{x}^\star\|_2$$

Combining these steps and using the tree-RIP condition $\delta_{3K} < 1/3$, we obtain the linear contraction $\alpha = 3\delta_{3K}/(1 - \delta_{3K}) < 1$ with noise term $\beta\|\boldsymbol{\varepsilon}\|_2$. □

B Backbone Ablation

Table 3 compares different backbone variants for the tree-aware proximal operator. Tree-ALISTA uses per-layer step sizes; Tree-ALISTA-MM adds momentum; the -Sym variants enforce symmetric weight matrices; Tree-HyperLISTA-Elem uses the full Tree-HyperLISTA backbone with element-wise thresholding (no tree structure).

Table 3: Backbone ablation: final NMSE (dB) at 16 layers on the core synthetic benchmark.

Backbone	**NMSE (dB)**	**#Params**
Tree-ALISTA	-15.00 ± 0.04	32
Tree-ALISTA-Sym	-13.63 ± 0.33	32
Tree-ALISTA-MM	-17.02 ± 0.03	32
Tree-ALISTA-MM-Sym	$+38.80 \pm 18.57$	32
Tree-HyperLISTA-Elem	-15.75 ± 7.06	3
Tree-HyperLISTA (M3)	$\mathbf{-28.66 \pm 0.10}$	**3**

The results show that the momentum-accelerated HyperLISTA backbone combined with tree-aware thresholding is critical. The symmetric + momentum variant (Tree-ALISTA-MM-Sym) diverges badly, confirming that the symmetric constraint is incompatible with momentum in this setting.

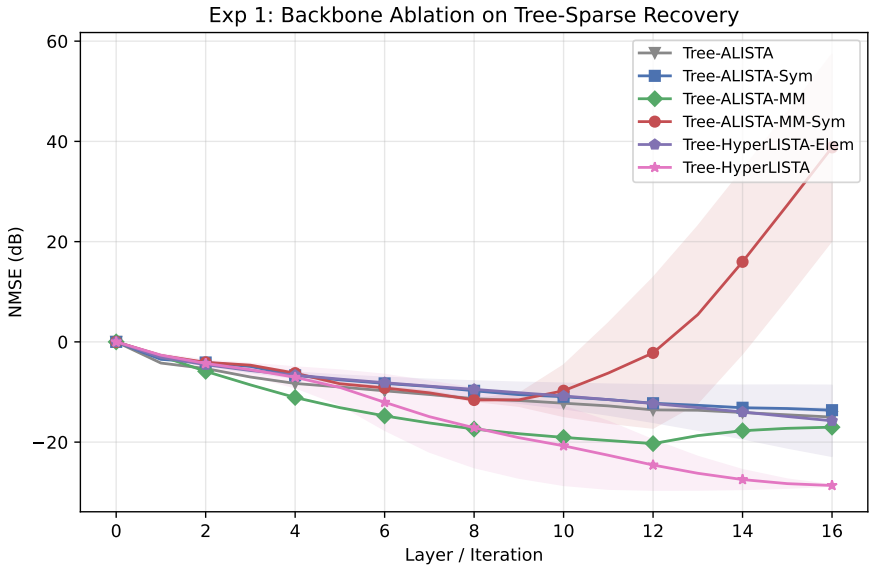


Figure 7: Convergence trajectories for backbone ablation variants.

Table 4: Extended mechanism ablation on the core benchmark.

Mechanism	**NMSE (dB)**
Tree Hard (M1)	-3.62
Tree Threshold (M2)	-2.73
Hybrid Tree (M3)	-2.73
Diff. Ancestor	-3.12
Gumbel Top-K	-2.73

C Extended Mechanism Ablation

Table 4 extends the mechanism comparison to include differentiable relaxations.

D Effect of ρ on Dense Signals

E Sparsity Mismatch

Figure 10 shows performance when the test-time sparsity ratio differs from training (ratio $= 1.0$). At ratio 0.5 (sparser than expected), TH-hard achieves -19.1 dB and Tree-CoSaMP achieves -22.2 dB. At ratio 1.5 (denser), all methods degrade similarly, suggesting sparsity overshoot is a universal challenge.

F Extended Mismatch Robustness

We evaluate mismatch along four additional axes: coefficient magnitude scaling, sensing matrix type, tree topology, and tree-consistency fraction.

Magnitude scaling. At scaling $\times 1.0$, TH-hybrid achieves -28.7 dB; it degrades gracefully to -28.1 dB at $\times 3.0$, far outperforming HyperLISTA which catastrophically diverges (> 85 dB NMSE).

Sensing matrix type. Under correlated, partial-orthogonal, and ill-conditioned sensing matrices, all methods perform similarly (NMSE $\approx$ 0.7–4.1 dB range), suggesting sensing matrix robustness is limited for all methods when the matrix is adversarial.

Figure 8: Extended mechanism comparison including differentiable relaxations.

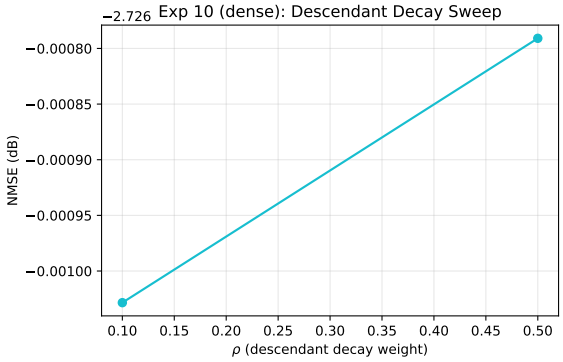


Figure 9: Effect of decay parameter ρ on denser signal regimes (ρ=0.1: -2.73 dB; ρ=0.5: -2.73 dB).

Tree topology. Under shuffled-parent and ternary-truncated trees, TH-hybrid achieves -11.4 and -13.6 dB respectively, compared to HyperLISTA's -18.1 and -18.4 dB (though HyperLISTA has high variance).

Tree consistency fraction. As the fraction of tree-inconsistent training examples increases from 0 to 0.5, TH-hybrid degrades from -28.8 dB to -7.2 dB, confirming that tree structure in the data is necessary for the tree-aware proximal to be effective.

G Low-SNR Regime

Figure 12 shows performance at SNR $\in \{10, 20, 30\}$ dB. TH-hybrid maintains its advantage at all SNR levels: -3.06 dB at 10 dB SNR, -3.86 dB at 20 dB, and -3.99 dB at 30 dB. Tree-CoSaMP benefits most from higher SNR ($-1.86 \rightarrow -9.20$ dB), while HyperLISTA plateaus early.

H Superlinear Convergence

Figure 13 shows that the per-instance tuned Tree-HyperLISTA (hyperparameters tuned to each test instance) achieves -4.0 dB at 6 layers, outperforming the globally-tuned variant (-3.3 dB). The gap motivates future work on instance-adaptive hyperparameter schedules.

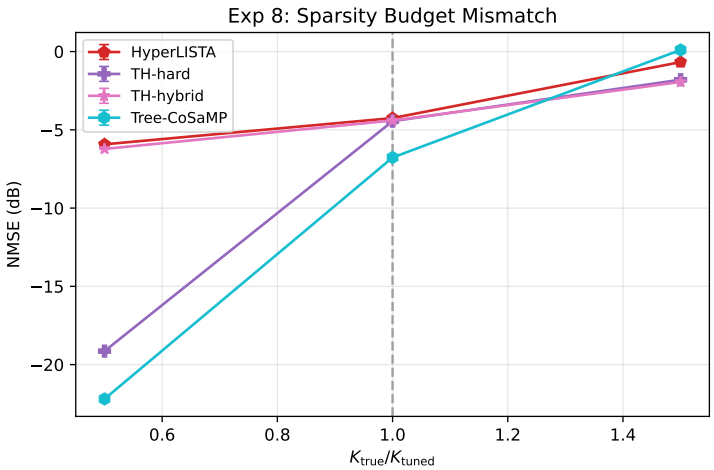
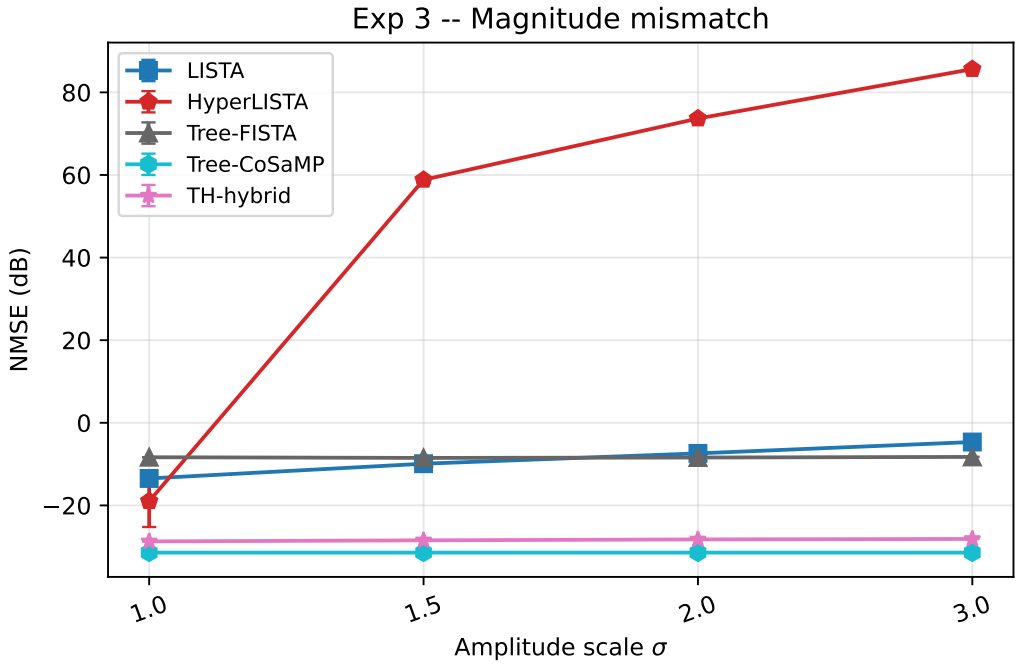
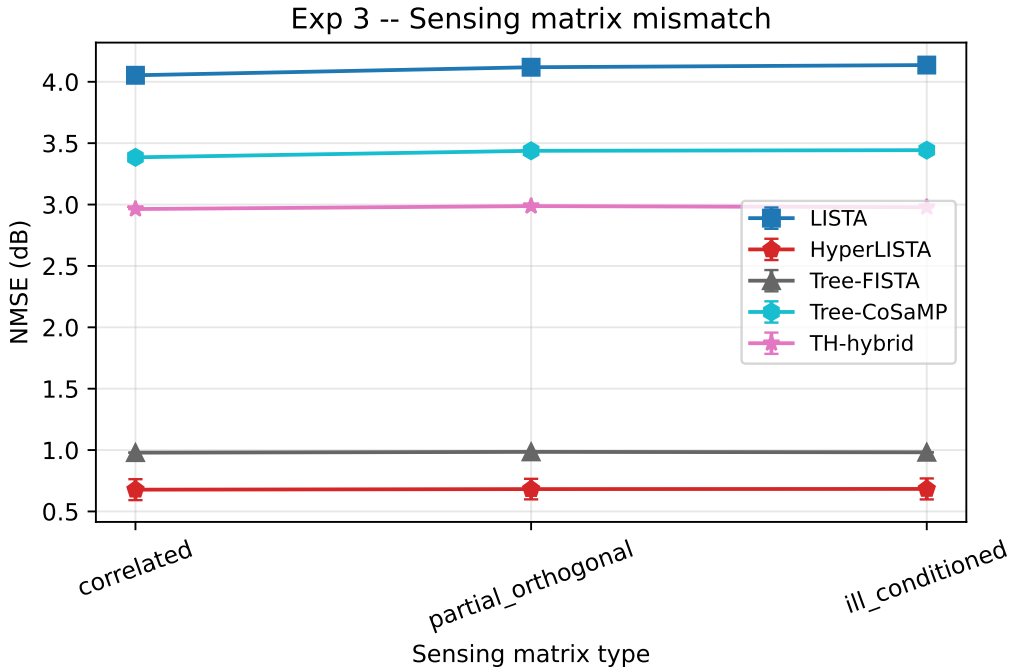


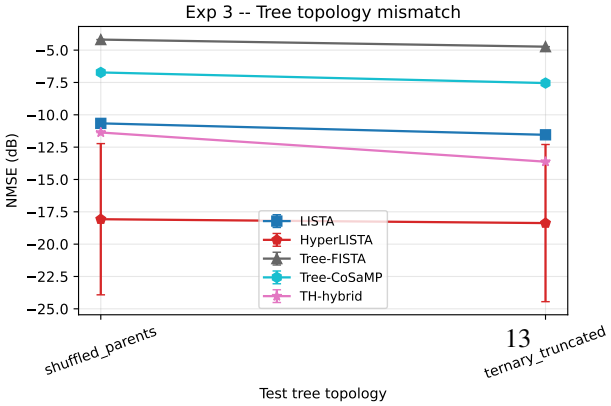
Figure 10: Sparsity mismatch: NMSE vs. sparsity ratio at test time (training ratio $= 1.0$).

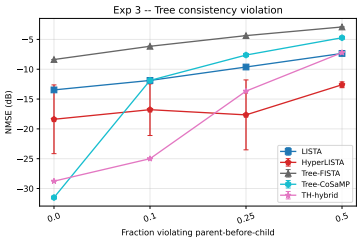
I Cross-Structure Experiments

Figure 14 evaluates Tree-HyperLISTA (trained on tree-sparse signals) vs. HyperLISTA (elementwise) on two sensing matrix types for both elementwise and tree-structured test signals. TH-hybrid achieves -3.45 dB on elementwise test data and -2.73 dB on tree-structured data, showing that tree-aware structure does not significantly hurt performance on non-tree signals.


Exp 3 -- Magnitude mismatch
LISTA
HyperLISTA
Tree-FISTA
Tree-CoSaMP
TH-hybrid
NMSE (dB)
80
60
40
20
0
−20
1.0
1.5
2.0
3.0
Amplitude scale σ


Exp 3 -- Sensing matrix mismatch
NMSE (dB)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
LISTA
HyperLISTA
Tree-FISTA
Tree-CoSaMP
TH-hybrid
correlated
partial_orthogonal
ill_conditioned
Sensing matrix type


Exp 3 -- Tree topology mismatch
NMSE (dB)
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
−20.0
−22.5
−25.0
LISTA
HyperLISTA
Tree-FISTA
Tree-CoSaMP
TH-hybrid
shuffled_parents
ternary_truncated
Test tree topology


Exp 3 -- Tree consistency violation
NMSE (dB)
−5
−10
−15
−20
−25
−30
LISTA
HyperLISTA
Tree-FISTA
Tree-CoSaMP
TH-hybrid
0.0
0.1
0.25
0.5
Fraction violating parent-before-child

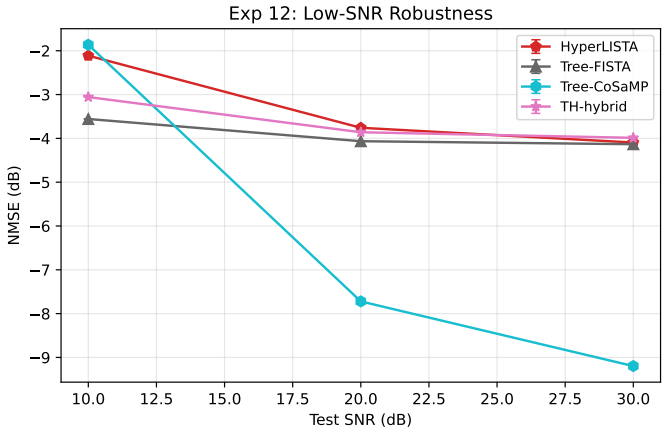


Figure 12: NMSE vs. SNR (dB) for Tree-HyperLISTA and baselines.

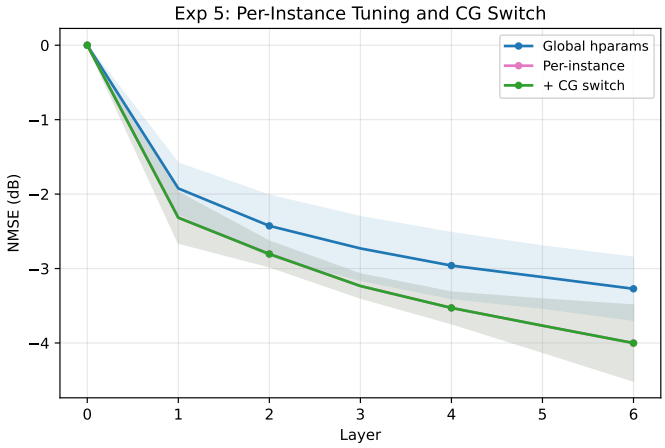


Figure 13: Superlinear convergence: global vs. per-instance and CG-switch variants.

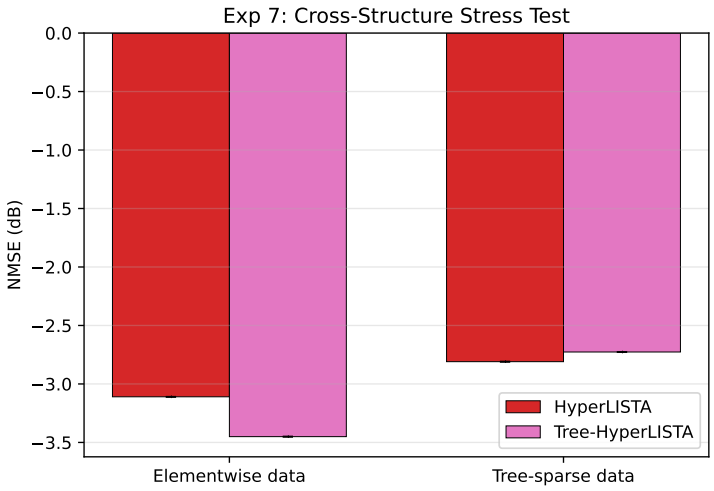


Figure 14: Cross-structure evaluation: tree-aware vs. elementwise models on both signal types.